# CaC-ConfigMgr: Operating Mode & Context

> **Fleet & Hierarchical Templates Demonstration**
> Centralized LogPoint Configuration Management in "Configuration as Code" Mode

## 🎯 Business Context

### The Problem

| Before | After with CaC |
|---|---|
| 50+ SIEMs managed manually | 1 codebase = 50+ deployments |
| Frequent human errors | Automatic validation before deployment |
| No change traceability | Git = complete history |
| Client onboarding = 2 weeks | Onboarding = 1 YAML file |
| Mass update = nightmare | 1 template change → all clients |

### Our MSSP Architecture

```
┌──────────────────────────────────────────────────────────────────────┐
│                        LOGPOINT (Vendor)                             │
│            ┌──────────────────────────────────────┐                  │
│            │      Golden Templates (Base)         │                  │
│            │   • Security standards               │                  │
│            │   • PCI-DSS, ISO27001 Compliance     │                  │
│            └──────────────────────────────────────┘                  │
└──────────────────────────────┬───────────────────────────────────────┘
                               │
                               ▼
┌──────────────────────────────────────────────────────────────────────┐
│                        ACME-CORP (MSSP)                              │
│            ┌──────────────────────────────────────┐                  │
│            │    MSSP Base + Addons/Profiles       │                  │
│            │   • Banking addon (MiFID)            │                  │
│            │   • Healthcare addon (HIPAA)         │                  │
│            │   • Profiles: simple/enterprise/...  │                  │
│            └──────────────────────────────────────┘                  │
└──────────────────────────────┬───────────────────────────────────────┘
                               │
                               ▼
┌──────────────────────────────────────────────────────────────────────┐
│                        CLIENTS (Instances)                           │
│      ┌──────────────┐   ┌──────────────┐   ┌──────────────┐         │
│      │    Bank A    │   │    Bank B    │   │    Corp X    │         │
│      │  (Premium)   │   │  (Standard)  │   │ (Enterprise) │         │
```

```
│     └──────────────┘   └──────────────┘   └──────────────┘      │
└──────────────────────────────────────────────────────────────────┘
```

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## 🏗 Template Architecture (6 Levels)

### Vertical vs Horizontal Inheritance

```
┌──────────────────────────────────────────────────────────────────┐
│                      TEMPLATE HIERARCHY                          │
├──────────────────────────────────────────────────────────────────┤
│                                                                  │
│  LEVEL 1: LogPoint Golden Base      ← Root, LogPoint standard    │
│           │                                                      │
│           ├───[HORIZONTAL]──▶ LEVEL 2: PCI-DSS Addon             │
│           │                            (card payment compliance) │
│           │                                                      │
│           └───[HORIZONTAL]──▶ LEVEL 2: ISO27001 Addon            │
│                                        (information security std)│
│                                                                  │
│  LEVEL 3: MSSP Base ──────────────────▶ extends PCI-DSS          │
│           │                            (VERTICAL inheritance)    │
│           │                                                      │
│           ├───[HORIZONTAL]──▶ LEVEL 4: Banking Addon             │
│           │                            (bank-specific)           │
│           │                                                      │
│           └───[HORIZONTAL]──▶ LEVEL 4: Healthcare Addon          │
│                                        (healthcare-specific)     │
│                                                                  │
│  LEVEL 5: Profiles ───────────────────▶ extend addons            │
│           │                            (VERTICAL inheritance)    │
│           ├─── simple        (light client)                      │
│           ├─── enterprise    (standard client)                   │
│           └─── banking-premium (premium bank)                    │
│                                                                  │
│  LEVEL 6: Concrete Instance ──────────▶ Specific client          │
│           │                            (VERTICAL inheritance)    │
│           └─── bank-a-prod   (code: BANKA, region: EU-WEST)      │
│                                                                  │
└──────────────────────────────────────────────────────────────────┘
```

### Concrete Example: Bank A Production

```
Complete inheritance chain:

┌──────────┬──────────────────────────┬──────────────────────────────────────────────
┐
│ Level    │ Template                 │ What it brings
│
```

```
  ├──────────┼────────────────────────┼──────────────────────────────────────────┤
  │ 1        │ golden-base            │ • 6 standard repos                       │
  │          │                        │ • Basic routing policies                 │
  │          │                        │ • Retention: 90 days                     │
  ├──────────┼────────────────────────┼──────────────────────────────────────────┤
  │ 2        │ golden-pci-dss         │ • +1 PCI-audit repo                      │
  │ (addon)  │                        │ • PCI retention: 7 years (2555d)         │
  │          │                        │ • PCI processing policies                │
  ├──────────┼────────────────────────┼──────────────────────────────────────────┤
  │ 3        │ acme-base (MSSP)       │ • Override retention: 90→180d            │
  │          │                        │ • Adds mount_warm, mount_cold            │
  │          │                        │ • Archive tiering                        │
  ├──────────┼────────────────────────┼──────────────────────────────────────────┤
  │ 4        │ acme-banking-addon     │ • +1 trading repo                        │
  │ (addon)  │                        │ • Banking retention: 10 years            │
  │          │                        │ • MiFID compliance                       │
  ├──────────┼────────────────────────┼──────────────────────────────────────────┤
  │ 5        │ acme-banking-premium   │ • MiFID enrichment                       │
  │ (profil)│                        │ • Trading logs normalization             │
  │          │                        │ • 4-tier storage (fast/warm/cold/nfs)    │
  ├──────────┼────────────────────────┼──────────────────────────────────────────┤
  │ 6        │ bank-a-prod            │ • client_code: "BANKA"                   │
  │ (inst.)  │                        │ • region: "EU-WEST"                      │
  │          │                        │ • Specific repo-secu override            │
  └──────────┴────────────────────────┴──────────────────────────────────────────┘
```

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## 🛰 Fleet Concept (Fleet Management)

### Definition

A **Fleet** = a group of LogPoint nodes (DataNodes, SearchHeads, AIOS) sharing the same configuration.

```yaml
# Fleet Bank A
spec:
  managementMode: director    # Managed via Director API
  director:
    poolUuid: pool-bank-a     # Isolated pool for the client
    apiHost: https://director.logpoint.com

  nodes:
    dataNodes:                # Log storage
      - name: dn-bank-a-01
        logpointId: lp-bank-a-p1
        tags: [cluster: production, env: prod, region: eu-west-1]
      - name: dn-bank-a-02
        logpointId: lp-bank-a-p2

    searchHeads:              # Search interface
      - name: sh-bank-a-01
        logpointId: lp-bank-a-s1
        tags: [cluster: frontend, env: prod]
```

### Node Types

| Type | Role | Example tags |
|---|---|---|
| **DataNode** | Storage and indexing | `cluster: production`, `retention: hot` |
| **SearchHead** | User interface | `cluster: frontend`, `role: primary` |
| **AIOS** | Analytics and ML | `cluster: analytics`, `gpu: enabled` |

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## ⚙ Operating Mode

### "Desired State" Workflow

```
┌──────────────┐     ┌──────────────┐     ┌──────────────┐
┌──────────────┐
│   YAML des   │────▶│   Template   │────▶│  Comparison  │────▶│
Application │
│   configs    │     │  Resolution  │     │ (plan/diff)  │     │   (apply)
│
└──────────────┘     └──────────────┘     └──────────────┘
└──────────────┘
       │                    │                    │                   │
       ▼                    ▼                    ▼                   ▼
```

```
    SIEM Team           6-level             See the             Automatic
    writes YAML         inheritance         changes             deployment
```

### Main Commands

```
# 1. VALIDATE - Check syntax and consistency
cac-configmgr validate demo-configs/
# → Check: YAML syntax, references, schemas

# 2. PLAN - Preview changes before applying
cac-configmgr plan \
  --fleet demo-configs/instances/banks/bank-a/prod/fleet.yaml \
  --topology demo-configs/instances/banks/bank-a/prod/instance.yaml \
  --templates-dir demo-configs/templates
# → Displays: resolved resources, variables, inheritance chain

# 3. APPLY - Deploy (when we reach Phase 2+)
# cac-configmgr apply --fleet ... --topology ... --auto-approve
```

### Client Configuration Example

```yaml
# instances/banks/bank-a/prod/instance.yaml
apiVersion: cac-configmgr.io/v1
kind: TopologyInstance
metadata:
  name: bank-a-prod
  extends: mssp/acme-corp/profiles/banking-premium  # ← Inherits premium
profile
  fleetRef: ./fleet.yaml                             # ← References fleet

spec:
  vars:
    client_code: BANKA        # ← Client-specific variable
    region: EU-WEST           # ← Client-specific variable

  # Specific overrides (optional)
  repos:
    - name: repo-secu
      hiddenrepopath:
        - _id: nfs-tier
          retention: 3650     # ← Override: 10 years for this client
```

---

## 🔄 Comparison: Bank A vs Bank B vs Corp X

### Same MSSP, Different Profiles

```
┌─────────────────────────────────────────────────────────────────────────┐
│                              BANK A                                     │
│   ┌─────────────────────────────────────────────────────────────────┐   │
│   │ Profile: banking-premium                                        │   │
│   │ Chain: base → pci-dss → mssp-base → banking → banking-premium   │   │
│   │                                                                 │   │
│   │ • 10 repos (incl. trading, pci-audit)                           │   │
│   │ • MiFID enrichment                                              │   │
│   │ • Max retention: 10 years                                       │   │
│   │ • Variables: client_code=BANKA, region=EU-WEST                  │   │
│   └─────────────────────────────────────────────────────────────────┘   │
└─────────────────────────────────────────────────────────────────────────┘

┌─────────────────────────────────────────────────────────────────────────┐
│                              BANK B                                     │
│   ┌─────────────────────────────────────────────────────────────────┐   │
│   │ Profile: banking (direct addon)                                 │   │
│   │ Chain: base → pci-dss → mssp-base → banking                     │   │
│   │                                                                 │   │
│   │ • 8 repos (no trading, no nfs-tier)                             │   │
│   │ • No MiFID enrichment                                           │   │
│   │ • Max retention: 7 years (PCI)                                  │   │
│   │ • Variables: client_code=BANKB, region=EU-EAST                  │   │
│   └─────────────────────────────────────────────────────────────────┘   │
└─────────────────────────────────────────────────────────────────────────┘

┌─────────────────────────────────────────────────────────────────────────┐
│                          CORP X (Enterprise)                            │
│   ┌─────────────────────────────────────────────────────────────────┐   │
│   │ Profile: enterprise                                             │   │
│   │ Chain: base → pci-dss → mssp-base → enterprise                  │   │
│   │                                                                 │   │
│   │ • 7 repos (standard + archive tiers)                            │   │
│   │ • No banking-specific                                           │   │
│   │ • Max retention: 7 years                                        │   │
│   │ • Variables: client_code=CORPX, region=US-EAST                  │   │
│   └─────────────────────────────────────────────────────────────────┘   │
└─────────────────────────────────────────────────────────────────────────┘
```

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## ✅ Key Benefits

1. **DRY Principle** (Don't Repeat Yourself)

```
× Before: 50 clients × 100 config lines = 5000 lines to maintain
✅ After: 6 templates + 50 instance files = ~800 lines only
```

2. **Simplified Mass Updates**

```
# Modify retention_default in acme-base
# → Impact: ALL ACME clients are automatically updated
```

3. **Fast Onboarding**

```
# New client = 1 YAML file
# Time: 5 minutes vs 2 weeks before
```

4. **Guaranteed Compliance**

```
PCI-DSS Addon → all banking clients inherit
ISO27001 Addon → all sensitive clients inherit
```

5. **Audit and Traceability**

```
Git log = complete change history
Who changed what when = immediately available
```

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## 📋 Demo Checklist

| Step | Command/Action | Objective |
|---|---|---|
| 1 | `tree demo-configs/ -L 4` | Show complete structure |
| 2 | `cac-configmgr validate demo-configs/` | Validate all configs |
| 3 | Plan Bank A | Show resolution + 6-level chain |
| 4 | Compare Bank A vs Bank B | Show profile flexibility |
| 5 | Show fleet.yaml | Explain fleet management |
| 6 | Highlight variables | client_code, region, retentions |

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## 🎤 Key Messages to Communicate

1. **"One codebase to manage 1 or 100 SIEMs with the same effort"**
2. **"6-level inheritance allows factoring 80% of the configuration"**
3. **"Validation before deployment = zero errors in production"**
4. **"Client onboarding in minutes, not weeks"**

5. **"Compliance by construction, not by verification"**

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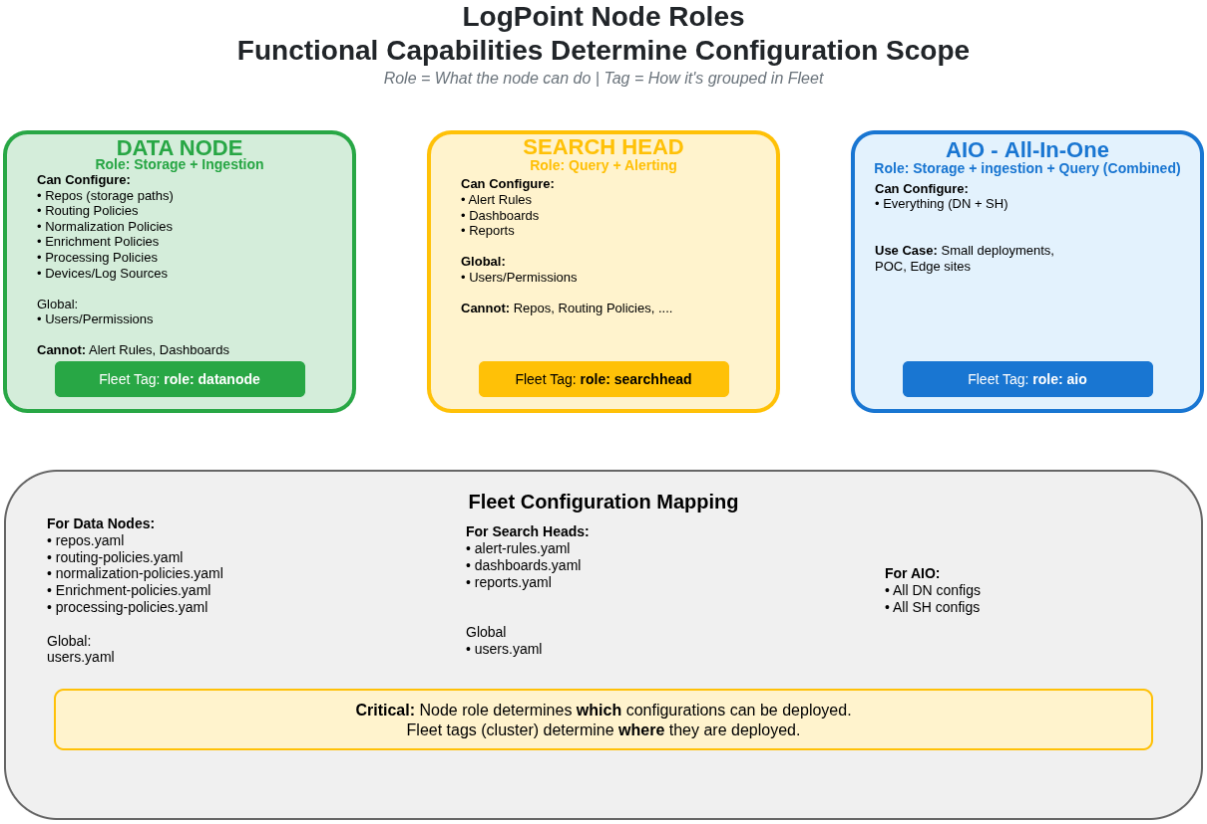
## 📎 Visual Diagrams

All diagrams are available in `docs/Presentations/TemplateHierarchy-Fleet/images/`.

### Fleet Architecture (01-fleet-architecture.drawio)

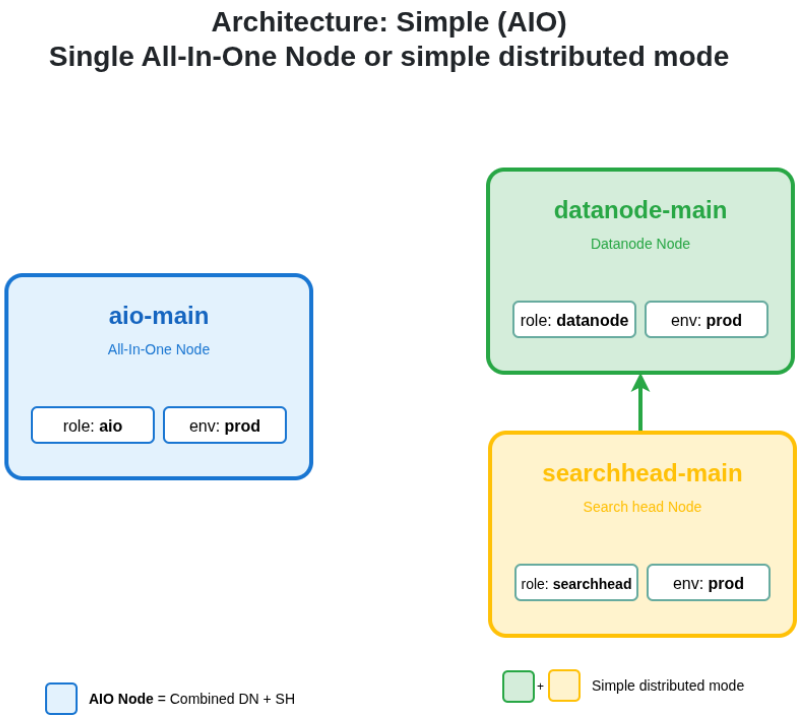
**Page 1: Node Roles**

*DataNode, SearchHead, AIO capabilities and fleet tags*



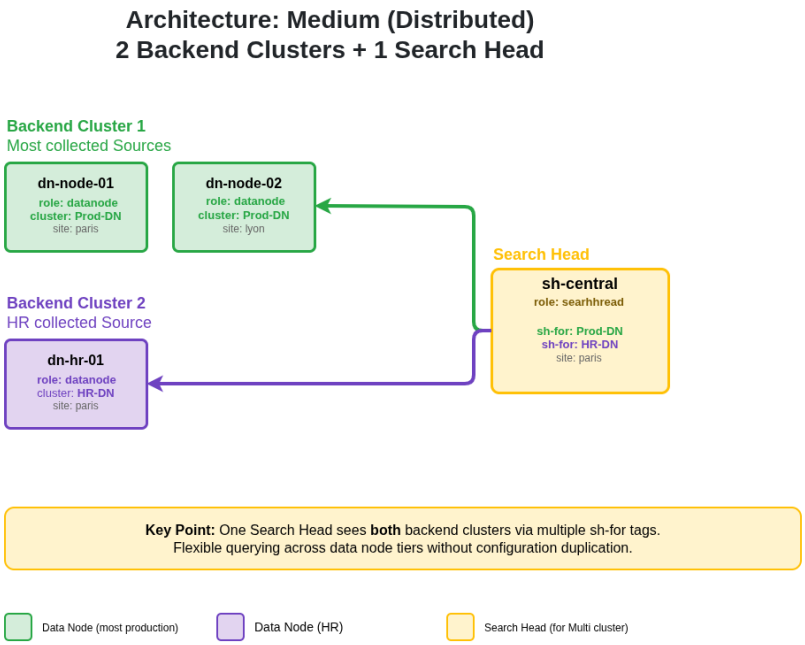

**Page 2: Simple Architecture**

*Single AIO or simple distributed mode*



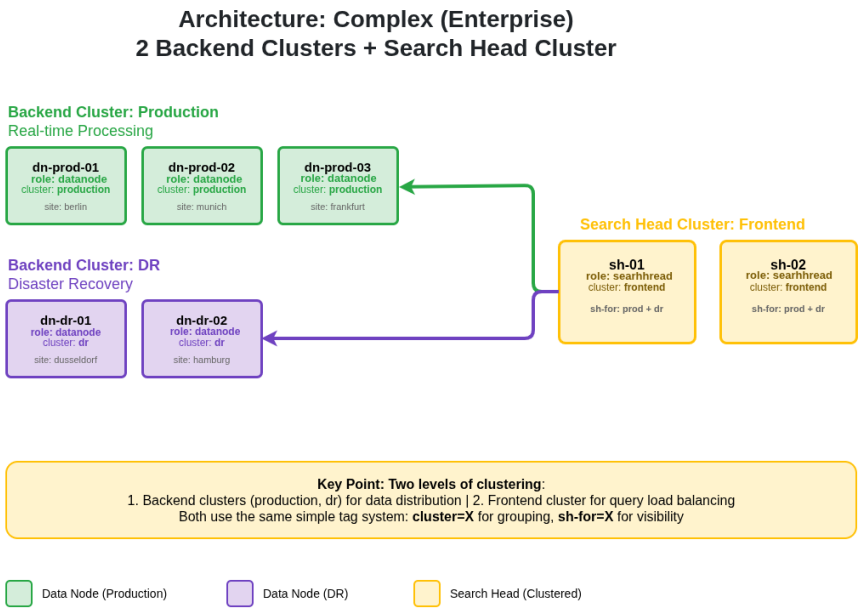

## Page 3: Medium Distributed

*2 backend clusters + 1 Search Head*




## Page 4: Complex Enterprise

*Production cluster + DR + SH cluster*

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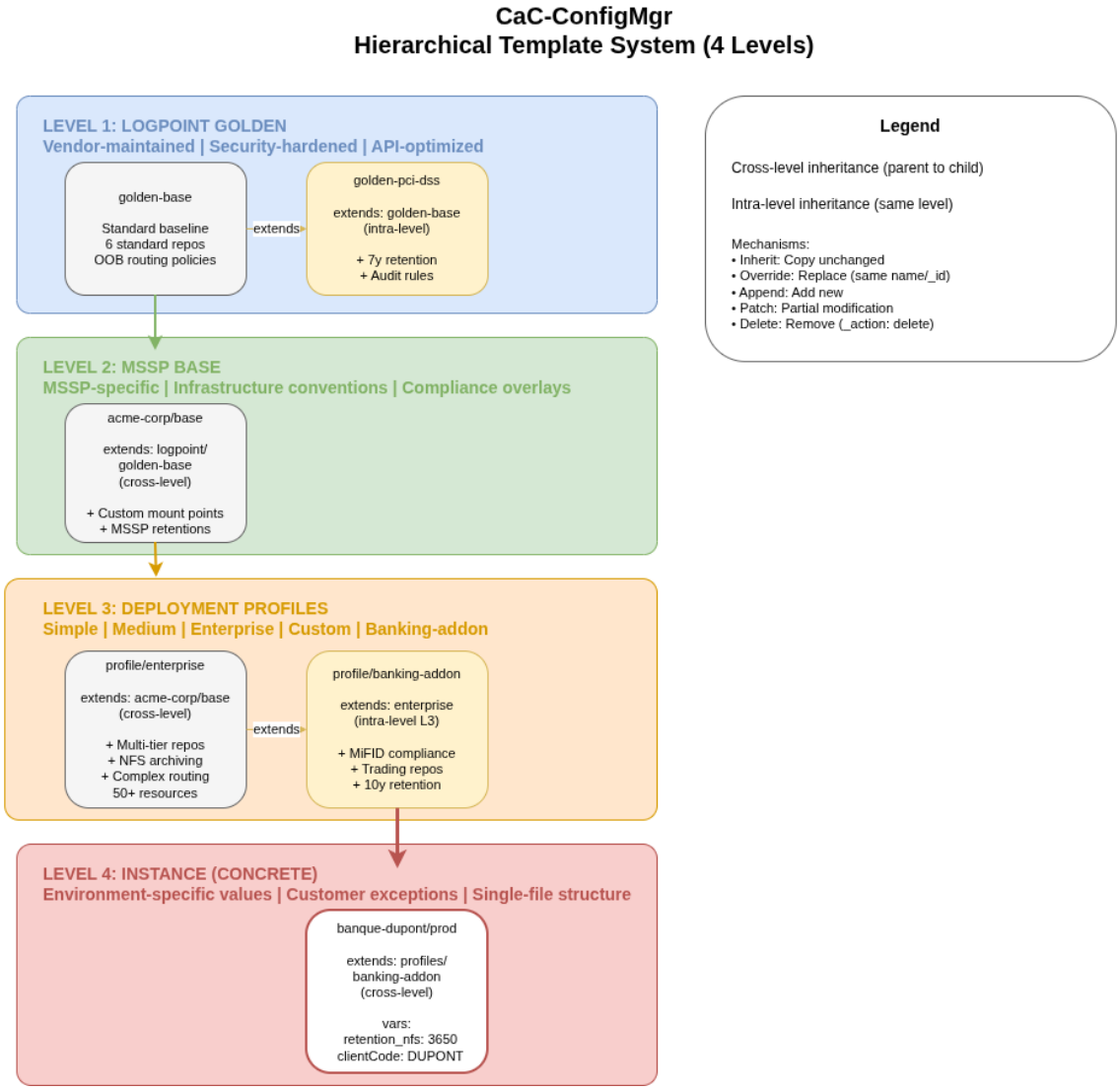
## Template Hierarchy (02-template-hierarchy.drawio)

### Page 1: 4-Level Overview

*Hierarchical template system (4 levels)*



## Page 2: Merge Mechanisms

*Deep merge with `_id` matching*

**Deep Merge and List Merging with _id**

**Parent Template**
**golden-base/repos.yaml**

```
repos:
  - name: repo-secu
    hiddenrepopath:
      - _id: fast-tier
        path: /opt/immune/storage
        retention: 365
      - _id: warm-tier
        path: /opt/immune/storage-warm
        retention: 90
```

deep_merge()

**Child Template**
**acme-enterprise/repos.yaml**

```
repos:
  - name: repo-secu
    hiddenrepopath:
      - _id: fast-tier
        retention: 7          <-- Override
      - _id: warm-tier
        retention: 30         <-- Override
      - _id: nfs-tier       <-- Append (new)
        path: /opt/immune/storage-nfs
        retention: 1095
```

**Final Resolution**
**(API Payload)**

```
{
  "repos": [{
    "name": "repo-secu",
    "hiddenrepopath": [
      {"path": "/opt/immune/storage", "retention": 7},
      {"path": "/opt/immune/storage-warm", "retention": 30},
      {"path": "/opt/immune/storage-nfs", "retention": 1095}
    ]
  }]
}

Note: _id fields are filtered before API call
```

**Ordering Directives**

```
_after: _id     -- Insert after element
_before: _id    -- Insert before element
_position: N    -- Absolute position (1-based)
_first: true    -- Force first position
_last: true     -- Force last position

Example:
- _id: crit-powershell
  _after: crit-verbose
  repo: repo-secu
  key: CommandLine
```

**Deletion (_action: delete)**

```
To remove an inherited element:

routing_criteria:
  - _id: crit-obsolete
    _action: delete
```

## Page 3: Complete Example

*Full repo inheritance chain*

**Deep Merge and List Merging with _id**

**Parent Template**
**golden-base/repos.yaml**

```
repos:
  - name: repo-secu
    hiddenrepopath:
      - _id: fast-tier
        path: /opt/immune/storage
        retention: 365
      - _id: warm-tier
        path: /opt/immune/storage-warm
        retention: 90
```

deep_merge()

**Child Template**
**acme-enterprise/repos.yaml**

```
repos:
  - name: repo-secu
    hiddenrepopath:
      - _id: fast-tier
        retention: 7          <-- Override
      - _id: warm-tier
        retention: 30         <-- Override
      - _id: nfs-tier       <-- Append (new)
        path: /opt/immune/storage-nfs
        retention: 1095
```

**Final Resolution**
**(API Payload)**

```
{
  "repos": [{
    "name": "repo-secu",
    "hiddenrepopath": [
      {"path": "/opt/immune/storage", "retention": 7},
      {"path": "/opt/immune/storage-warm", "retention": 30},
      {"path": "/opt/immune/storage-nfs", "retention": 1095}
    ]
  }]
}

Note: _id fields are filtered before API call
```

**Ordering Directives**

```
_after: _id     -- Insert after element
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_last: true     -- Force last position

Example:
- _id: crit-powershell
  _after: crit-verbose
  repo: repo-secu
  key: CommandLine
```

**Deletion (_action: delete)**

```
To remove an inherited element:

routing_criteria:
  - _id: crit-obsolete
    _action: delete
```

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## Export Instructions

If you need to re-export from draw.io:

**Files to open:**

- `docs/Presentations/TemplateHierarchy-Fleet/schemas/01-fleet-architecture.drawio`
- `docs/Presentations/TemplateHierarchy-Fleet/schemas/02-template-hierarchy.drawio`

**Export settings:**

- Format: PNG
- Border: 10px
- Crop: Crop to Content
- Save to: `docs/Presentations/TemplateHierarchy-Fleet/images/`

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*Document prepared for CaC-ConfigMgr demonstration*
*Phase 1: Foundation (Template Resolution) | Phase 2: Director Integration (Plan/Apply)*